

REPRESENTATION AND “RELIABLE PRESENCE”

Summary. The “New Computationalism” that is the subject of this special issue requires an appropriate notion of representation. The purpose of this essay is to recommend such a notion. In cognitive science generally, there have been two primary candidates for spelling out what it is to be a representation: teleological accounts and accounts based on “decoupling.” I argue that the latter sort of account has two serious problems. First, it is multiply ambiguous; second, it is revisionist and alienating to many of the potential allies of the “New Computationalism”. I also suggest that teleological accounts do not suffer from these problems, making them more appropriate as the foundation of any new computationalism.

If computationalism, the idea that (at least some of) human thought is computation, is to be the basis for cognitive science, then it must be an empirical matter whether (at least some of) human thought is computational. This, I take it, is an obvious necessary condition for anything to be a scientific hypothesis. This obvious necessary condition rules out immediately understandings of computation like the one described by Agre in his *Computation and Human Experience*. There Agre writes, “For my purposes, *computation* relates to the analysis and synthesis of especially complicated things. These analytic and synthetic practices are best understood as nothing less grand or more specific than an inquiry into physical realization as such.” (1997, p. 11) According to this understanding of computers, it is a necessary truth that human thought is not computation. Erring in exactly the opposite direction is Brian Cantwell Smith, for whom computation is not a distinctive subject matter and computers are just intentional entities of middling complexity. For Smith it seems as if it is necessarily true that human thought is computation, because everything intentional is (see Smith 1996, Smith forthcoming). Thus it seems that neither Agre nor Smith provide an appropriate understanding of computation to ground the “New Computationalism” that is the topic of this journal.

Another, perhaps more appropriate understanding of computation takes computation to be the rule-governed manipulation of representations. This way of thinking of computation, typical of philosophers, is nicely exemplified in Jerry Fodor’s computational theory of the mind (see Fodor 1981), where computation is thought of as *formal symbol manipulation*, as well as John Haugeland’s explanation of computers as *interpreted automatic formal systems* (see Haugeland 1985). In both these understandings of computation, there are representations which undergo rule-governed changes. These representations have both formal and semantic properties; the formal properties are the ones manipulated, as the semantic properties are not causally-potent. Though there are interesting differences between these two understandings of

computation,¹ one thing on which they both agree is that computation is necessarily representational.

Assuming that computation is necessarily representational, recent challenges to the role of representation in cognitive science should also be seen as challenges to the role of computation in cognitive science. van Gelder, for example, takes his dynamical hypothesis (1995, 1998) to present a challenge to computationalism, in part at least because it challenges representationalism. This anti-representationalist challenge to computationalism has changed the tenor of recent writings on representation: the debate has changed, in part at least, from being about how to determine the content of representations to a debate about *what it is to be a representation* in the first place. It is this question that will be my focus here.

There have been two primary types of account of what it is to be a representation in this debate: teleological definitions and definitions based on “decoupling.” In this paper, I will present some difficulties and confusions concerning the latter type of account of what it is to be a representation. These problems make the associated family of definitions of representation inappropriate as a basis for a “New Computationalism”. I will also suggest that the teleological definition does not suffer these confusions and difficulties, and is therefore a better way to think about representation.

The family of definitions I will be criticizing, those based on “decoupling,” derive from John Haugeland’s “Representational Genera” (1991). Haugeland’s essay has become the touchstone for discussions of representational in cognitive science, despite the fact that only two of its twenty-seven pages are devoted to discussing representation-in-general. Haugeland’s definition, which is in all essentials identical to that of Brian Cantwell Smith (1996),² is nicely summarized by Andy Clark³ (1997) as follows. A system counts as *representation using* just in case:

It must coordinate its behaviors with environmental features that are not always “reliably present to the system.”

¹ Surprisingly, Haugeland has exhibited a systematic confusion here, taking his and Fodor’s understandings of computation to be the same. In fact, however, they differ very significantly. In Haugeland’s interpreted automatic formal systems, for the system to be representational, and hence computational, its behavior must be *rationaly interpretable*. That is, for any of a system’s states to be representation, the system’s overall behavior must be coherently interpretable. For Fodor, the fact that a system’s states are representations has nothing whatsoever to do with interpretation. For Fodor, content is intrinsic to the states, and has nothing to do with behavior. In fact, Fodor has always been very vocal about his impatience with interpretationalism (see Fodor 1985). Thus Haugeland and Fodor take the representational status of computational states to come from entirely different sources. That Haugeland makes this error is evident in his “The Intentionality All-Stars” (1990), in which he explicitly places Fodor at “first base”, and ascribes interpretationalism to him.

² Smith actually talks about what he calls ‘registration’, a theory of representation plus ontology. If we de-ontologize the theory--taking it to be just a theory of representation--it is exactly the same as Haugeland’s.

³ It is worth pointing out here that I will be picking on Andy Clark quite a bit in what follows. I do so only because he has written more, and more clearly, on this topic than anyone else.

It copes with such cases by having something else “stand in” for those features and guide behavior.

The “something else” is part of a more general representational scheme that allows the standing in to occur systematically and allows for a variety of related states. (Clark, 1997, p,144; see also Haugeland, 1991, pp.62-3)

A *representation*, according to this definition, is something that acts as a stand-in in such a system. This, as it stands, is hopelessly fuzzy. The phrase “reliably present to the system” is left unanalyzed by Haugeland, who only intended this to be “a few dogmatic and sketchy remarks.”(p.62)⁴ In the remainder of this essay, I will argue that there is no one way to spell out “not always reliably present to the system.” In fact, it seems that the phrase is ambiguous among three possibilities, ruling out only those cases in which the representation in the system is constantly causally coupled with the thing represented. This ambiguity has been the cause of considerable confusion. We would be better off, I will argue, if we do not define representation in terms of reliable presence, or lack thereof.

In order to make this case I will need to introduce the following three terms:

- A representation R and its target T are in *constant causal contact* just in case whenever R is present in a system, T is causing it.
- A target T is *absent* just in case T has no local causal effects when a representation R of it is present in a system. (Both my grandmother and the number 3 are absent in this sense.)
- A representation R is *adaptive* at time t just in case R’s presence in a system at t is neither the result of a malfunction of any kind, nor leads the system to behave inappropriately with respect to its surroundings at time t.

Wheeler (1996), who like van Gelder and Haugeland intends only a rough characterization, glosses representation as follows:

Consider an agent A, one of A’s internal states, S, and some feature F, of A’s environment (where F may be an aspect of A’s surroundings, or of A’s own activity). A, S, and F are all picked out by some external observer, O. O correctly identifies S to be a representation of F if, on those occasions when O’s best explanation of A’s behavior is that A is coordinating its behavior reliably with F,

1. S is the cause of A’s behavior, an

2. S has been previously (or is now) a cause of A’s behavior on an occasion when F was (is) not actually present to A, but--according to O’s best theory--A was (is) still coordinating its behaviors with respect to F. (pp.214-5)

⁴ It is rather ironic that Haugeland’s “dogmatic and sketchy remarks” have been taken by so many to amount to a definition of representation. In the 1991 paper, Haugeland spends just over a page on defining representation; the paper is 28 pages long.

Wheeler thus offers (in his second criterion) a somewhat more specified parallel to the Haugeland/Smith requirement that the thing represented need not be reliably present: the representation must cause behavior coordinated with the represented when that represented is absent—that is, distant in space or time—at least once. Wheeler can be read in two ways, depending on whether ‘coordinate’ is read as a success verb or not. If coordinate is taken to be a success verb, the definition is the same as those we will consider below. On another reading, in which coordinate is read “thinly”, Wheeler’s definition would be satisfied by a system that occasionally *misrepresents*. Misrepresenting is not simply representing things that are not the case—if it were, desires, which are paradigmatically for things which are not the case, would always be misrepresentations. So misrepresentation must involve more than simple falsehood. Instead, in order to misrepresent, a system must produce a representation that something is the case, when that something is not in fact the case, and it must do so in a *non-adaptive* way, such that it is the result of some mistake or malfunction in the system. Misrepresentation is one possible gloss of “not always reliably present to the system”. Wheeler on this second reading of his definition, provides us with “not always reliably present #1”: R and T **are not** in constant causal contact, T **is** absent, R **is not** adaptive.⁵

But in *Being There* (1997), which we might call “The Early Clark”, Andy Clark claims that the Haugeland/Smith definition requires what he (Clark) calls *decouplability*, which he defines as “the capacity to use the inner states to guide behavior in the absence of the environmental feature [represented].”(1997, p.144) This, on the face of it, seems to require no more than Wheeler’s definition; that is, the decouplability requirement seems as if it would be satisfied by a system that sometimes misrepresents. Clark apparently does not see it this way, though. He seems to read Haugeland’s “not always reliably present” as requiring high-level reasoning, the ability to solve what he calls “representation-hungry problems”.

Adaptive hookup thus phases gradually into genuine internal representation as the hookup’s complexity and systematicity increase. At the far end of this continuum we find Haugeland’s creatures that can deploy the inner codes in the total absence of their target environmental features. Such creatures are the most obvious representers of their world, and are the ones able to engage in complex imaginings, off-line reflection and counterfactual reasoning. Problems that require such capacities for their solution are representation-hungry, in that they seem to cry out for the use of

⁵ Ruth Millikan and John Haugeland have both argued that the correct way to interpret Wheeler’s definition is with ‘coordinate’ taken to be a success verb. They argue that, given this, occasional misrepresentation is not sufficient to satisfy Wheeler’s definition. Against this, it is worth pointing out that the examples that Wheeler cites in his essay of systems that fail to meet his definition, and so have no representations, are *guaranteed* to succeed at their task every time. In other words, Wheeler’s paradigm cases of anti-representational systems never behave inappropriately, given their task; they don’t represent, it seems, because they never misrepresent.

Very little rides on a correct interpretation of Wheeler’s definition. If Millikan and Haugeland are right and I am wrong here, Wheeler’s definition will still have the other problems that those of Clark, Grush, Haugeland, and Smith do.

*inner systemic features as stand-ins for external states of affairs. (1997, p.147)*⁶

Decouplability in the Haugeland/Smith sense, then, is “not always reliably present #2”. It requires that R and T **are not** in constant causal contact, that T **is** absent, and that R **is** adaptive. Clark *rejects* decouplability as a necessary condition for something to be a representation. Using a neural group in the posterior parietal cortex of rats as an example, Clark argues that there are systems that are usefully called representational that can never be decoupled from the things about which they carry information. This complex of neurons, he says, carries information about the position of the rat’s head, but there is no reason to think that “these neurons can play their role in the absence of a constant stream of proprioceptive signals from the rat’s body.” (p. 145) So, according to the Early Clark (in *Being There*), something can be a representation when it is always coupled to the thing it represents: a representation, that is, *might be* in constant causal contact with what it represents.⁷

In more recent papers (Clark 1997a; Clark and Grush forthcoming), however, Clark has made some moves toward reclaiming decouplability. In these papers, a distinction is made between *weak* and *strong* internal representation. A *weak* internal representation is what Clark calls a representation in *Being There*:

First, [representational mechanisms] must involve inner states whose functional role is to coordinate activity with its world (no mere correlations). Second, we must be able to identify specific inner states or processes with specific representational roles-- we must be able to isolate, within the system, the inner parameters or processes that stand-in for particular extra-neural states of affairs (otherwise we confront only complex inner state implicated in implicated in successful agent-environment coordination). (Clark 1997a, p.465)

To be a *strong* internal representation (what Clark and Grush [forthcoming] call an example of *minimally robust* representation), a system or entity must meet a third criterion, in addition to the two above.

And lastly, the system must be capable of using these inner states or processes to

⁶ Haugeland repudiates this as an interpretation of his brief discussion of representation (in his review of this essay), claiming that it is quite clear that this is not what he intended. Even if this is not what Haugeland intended, it is the way his definition has been interpreted by those who use it in discussing anti-representationalism. (In addition to those mentioned in the text, see DeGroot and Haselager, unpublished and Chemero 1998.) Furthermore, this unquestionably *is* what Brian Cantwell Smith, who takes himself to agree with Haugeland, intends. See Smith 1996, Chapters 6 and 7 (and especially p.241).

⁷ Indeed, here Clark gestures toward exactly the type of understanding of representation that I will eventually recommend. See Clark 1997, p.146. Van Gelder (1995) seems to agree. In “What Might Cognition be, if not Computation?” (1995), van Gelder, who cites Haugeland’s remarks as a source, says that any reasonable characterization of representation will be “based around a core idea of some state of a system, which in virtue of some general representational scheme, stands in for some further state of affairs, thereby enabling the system to behave appropriately with respect to that state of affairs.” This leaves Haugeland’s “not always ‘reliably present to the system’” unmentioned, presumably because it was deemed too exclusive.

solve problems off-line, to engage in vicarious explorations of a domain, and so on. It is this last capacity that distinguishes model-using agents from the rest. Strong internal representation is thus of a piece with the capacity to use inner models instead of real-world action and search. (Clark 1997a, p.465)

So strong internal representation does require decouplability. In more recent papers, though, Clark follows Grush (1995, 1997) in attempting to illustrate what he takes decouplability to be in terms of *emulation*. We can call Clark's emulator phase "the Later Clark".

An emulator is a mechanism within a system that takes information about the current state of the system and gives a prediction of the next state of the system as output (Grush, 1995, 1997; Clark and Grush forthcoming). Consider (their example) skilled reaching. Moving an arm and hand toward some object depends upon the brain receiving and responding to a stream of visual and proprioceptive feedback concerning the position and trajectory of the arm and hand. But often, due to the inherent speed limitations of the nervous system, the feedback is required faster than it is available. It is in situations like this that emulation is crucial. An emulator in such a case could take as input the current position of the arm and hand, along with the direction of their movement, and provide a sort of mock feedback as output, predicting the position and trajectory of the arm before the actual feedback arrives (Clark and Grush forthcoming, ms, pp. 4-5). Emulation, Clark and Grush claim, is important because it is the minimal case of strong internal representation, representation that is decouplable in the Haugeland/Smith sense:

In sum, it is our suggestion that a creature uses full-blooded internal representations if and only if it is possible to identify within the system specific states and/or processes whose functional role is to act as decouplable surrogates for specifiable, usually extra-neural states of affairs. Motor emulation circuitry, we think, provides a clear, minimal and evolutionarily plausible case in which these conditions may be met. (Clark and Grush forthcoming, ms, p.9)

There is some question, however, whether simple cases of emulation really are cases of decoupling in the Haugeland/Smith sense, requiring that the target be potentially *absent*. There is one sense in which we might say that an emulator controlling skilled reaching is decoupled: in the short time between when it takes its input (the state of the arm, the direction of its motion) and when it gives its output (mock feedback to guide action), the emulator is not receiving input directly from what it is representing; that is, unlike the locus of the actual proprioceptive feedback (which is a representation only in the Later Clark's [and Clark and Grush's] weak sense), the emulator's hookup with the target is not constant. This however is *not* decoupling in Clark's original sense: the capacity to use the inner states to guide behavior *in the absence of* the environmental feature represented. Despite the fact that the emulator is not hooked-up to incoming proprioceptive signals for a few milliseconds, the arm and the action it is undertaking is in no way *absent*. With some emulators, that is, R and T **are not** in constant causal contact, T **is not** absent, and R **is** adaptive. But Haugeland/Smith decouplability requires that R and T **are not** in constant causal contact, that T **is** absent, and that R **is** adaptive. Emulation, then, is "not always reliably present #3": R and T **are not** in constant causal

contact, T **is not** absent, R **is** adaptive. So emulation is not sufficient for decouplability; thus emulation is not sufficient for Haugeland's requirement that the thing represented need not be always reliably present (at least on Clark's strong reading of it--as requiring "the total absence of the target"). So, as depicted in figure 1, claiming that it is a necessary condition on something being a representation that what it represents is "not always reliably present to the system" is ambiguous among three possibilities:

- It might mean that the system sometimes misrepresents (which would make a comparatively large number of systems representation users);
- It might mean that the system contains emulators (yielding fewer representations users);
- It might mean that the system is capable of solving representation-hungry problems (allowing fewer still representation users).

The distinction between emulation and strong decouplability (in the Haugeland/Smith sense) is not a distinction without a difference. The distinction (and the difference) can be illustrated with examples from *coupled oscillator theory*.⁸ Start by considering that Clark and Grush (forthcoming) provide as an example of emulation a structure in the mid-brain of a cat which mimics some of the dynamic parameters of cat forelimb movement. With the use of such an emulator, the cat has access to *current* limb position information even though the proprioceptive signal coming from the limb's reports of *past* limb position information. Such an emulator can be implemented with a modification of a *simple relaxation oscillator*, such as those that have been used to model the behavior of neurons (Fitzhugh 1961). When presented with an input pattern consisting of voltage pulses, a relaxation oscillator will synchronize its firing with the pulses. If these pulses are rhythmical, the oscillator synchronizes and "beats along" by virtue of emitting its own pulses in tandem. A connected group of these oscillators can couple with rhythmical input patterns in ways which mirror the *metrical structure* of the patterns (Eck, Gasser and Port, 1999). That is, a network can distinguish weak beats from strong beats and can even distinguish *rests* using appropriate inhibitory connections. The main thing to notice about relaxation oscillators is that when the driving stimulus is removed from one, it relaxes immediately and returns to a quiescent state with no relation to the regularities in the input. That is, the oscillators are unable to decouple from their inputs. In fact, they simply respond to whatever energy comes to them via constant causal contact. That is, they respond at time *t* only to the input presented to them at time *t*.

The emulator governing cat forelimb motion discussed by Clark and Grush can be built with a simple, and biologically plausible (see Huber 1998), modification of a network of relaxation oscillators. To do so, one need only implement time delays so that the network can use the "stale" information from the proprioceptive stream

⁸ For more references and a more complete discussion of coupled oscillators in cognitive science, see Chemero and Eck 1999. The discussion of oscillators as representational vehicles draws heavily on that paper.

appropriately. Given that cat locomotion is periodic, such a delay could be found simply by measuring the rate of change in the proprioceptive stream and using it to estimate how fast the legs are moving. Once the delay is found, the emulator is able to predict current proprioceptive information using information from exactly one leg cycle before. The addition of time delays on the “axons” of the artificial neurons is both simple to implement and neurally plausible.

The stronger, Haugeland/Smith version of “not necessarily reliably present”—the -CCC, +Adaptive, +Absent case—cannot be modeled with *any* version of relaxation oscillators. As soon as the input to such an oscillator subsides, it relaxes, so it cannot be used to stand-in for anything absent. The +Absent case can, however, be implemented with *adaptive oscillators* (McCauley 1995). Adaptive oscillators, like relaxation oscillators, synchronize themselves quickly with rhythmical input, but, unlike relaxation oscillators, they have *momentum* and can maintain their oscillation after the cessation of the input stimulus. Such oscillators can beat along in real time to rhythmical stimuli, a task akin to tapping one's foot along with music. But after adaptive oscillators bring themselves into time with the input, they will continue to “expect” beats for a significant period *after the input has ceased*. This is akin to continuing to tap your feet along with the beat for a few minutes after your favorite song has ended. (That is, when it is +Absent.) It is crucial to note that the adaptive oscillators required to implement the Haugeland/Smith variety of decoupling are very different from the slightly modified relaxation oscillators that can implement emulators. So far from being a simple implementation of Haugeland/Smith-style decoupling, emulators are really not significantly different from oscillators require constant causal coupling.

Given the close similarity of oscillators that act as emulators and those that are constantly coupled, it is hard to tell what makes emulators theoretically interesting. Grush claims (1995, 1997) that the interesting and important thing about emulators is that emulation is a genuine case of representation, as opposed to the “mere presentation” we get in what he and the Later Clark (in Clark and Grush forthcoming) call weak representation. Indeed, he makes rather a big deal out of this, claiming that failing to distinguish between mere presentations and full-blooded representations has led everyone astray.

Most of the recent philosophical literature on internal representation has been tying itself in knots because it has not distinguished internal representations from internal presentations, and has been trying futilely to give a theory of content for internal presentations, and then subjecting these theories to constraints and intuitions which properly belong to representations. (Grush, 1997, p. 7)

The implication here is that before recent work in the philosophy of mind (Grush cites Dretske, Fodor and Millikan as the main culprits), we had intuitions concerning representations in his “minimally robust” sense, and they were not also intuitions about so-called mere presentations.⁹ The fact is, however, that if such carefree days ever

⁹ Brian Cantwell Smith may also be guilty of this. “This overall picture allows representation to return to its etymological roots—to be *re*-presentation, to involve ‘presenting again.’” (Smith 1996, p.351)

existed, it was long before there was cognitive science. The term ‘representation’ in Anglo-American philosophy and, later, in cognitive science comes from translations of Kant’s term *Vorstellung*. Notice, first of all, that *Vorstellung*, which literally means ‘a placing-before,’ would be better rendered into English as ‘presentation,’ and not as ‘representation,’ as the tradition has it. More importantly, the Kantian term *Vorstellung* is considerably more vague than the English ‘representation.’ W.H. Werkmeister points to sections of Kant’s first *Critique* in which *Vorstellung* demands translation as each of ‘cognition,’ ‘image,’ ‘representation,’ ‘conception,’ ‘sensation,’ ‘image,’ ‘concept,’ ‘notion,’ ‘perception,’ and ‘idea’ (Werkmeister, 1982, p.141). Furthermore, it is not just Fodor, Dretske and Millikan who use the term representation in this traditional, less restrictive sense: so too do cognitive scientists of all stripes (see for example Marr 1982, Miller and Freyd 1993, Jackendoff 1994) and cognitive neuroscientists (e.g. Kosslyn 1994, Churchland and Sejnowski 1992). So the fact that our ‘representation’ is a generic term, whose species include both weak and strong internal representation (in Clark and Grush’s sense), should not be surprising at all; it has never been restricted to the strong (that is, necessarily decoupled) type.

Defining representation in terms of decoupling, as Haugeland, Smith, (the Later) Clark and Grush wish to, is therefore *revisionist*. This in itself is not necessarily a bad thing: it is frequently advisable, even necessary to restrict the extensions of terms in the interest of clarity. And representation might be a term in need of restricting. But restricting ‘representation’ to the necessarily uncoupled cases goes too far for two reasons: it guarantees that all of the models in the so-called “embodied cognitive science” movement are non-representational, and hence non-computational. This includes connectionist networks, dynamical systems models, and many neuroscientific models. If these models of cognition are barred from taking part in the “New Computationalism” that this conference seeks to define in virtue of being non-representational, it is hard to imagine whom among cognitive scientists will be counted as allies. Furthermore, there are no neurally plausible models of the stronger Haugeland/Smith version of the definition. If the “New Computationalism” can point only to GOFAI (Haugeland 1985) and other neurally far-fetched models for examples, I can see no reason to call it new. These revisionist, restrictive definitions of representation are, therefore, inappropriate for our purposes here.¹⁰

We have seen two distinct problems with defining representation in terms of the represented not being always reliably present. First, those who try to define representation in terms of “reliable presence” do not agree about what that phrase means: is it the possibility of misrepresentation, emulation, or strong decouplability,? Secondly,

¹⁰ Notice too that this revisionism changes the nature of claims made by anti-representationalists, making their case both easier and less radical than, I think, they wish it to be. van Gelder (1995), Wheeler (1996), and Ramsey (1997) for example clearly intend to be arguing against representation of any sort. Carrying the day by changing the meaning of the term at issue would be at best a Pyrrhic victory for anti-representationalists, significantly de-radicalizing their claims. The revisionist, restrictive definition is also not appropriate for analyzing anti-representationalist claims.

we have seen that defining representation in terms of strong decouplability is *revisionist*, it changes the meaning of the term. It does not get us back to the original meaning of representation: the term representation has always covered both “mere presentations” and “full-blooded representations.” The conclusion to be drawn from this, I suggest, is that we should not define representation in terms of decoupling or lack of reliable presence. And this is, in fact, the key difference between the definitions of Haugeland, Wheeler, Clark,¹¹ Grush, as well as all the informational and causal theorists on one hand, and the teleological definition that I, along with Millikan and Bechtel,¹² recommend, on the other.

Such a teleological definition might be spelled out as follows:

A feature R0 of a system S will be counted as a *Representation for S* if and only if:

- (R1) R0 stands between a representation producer P and a representation consumer C that have been standardized to fit one another.
- (R2) R0 has as its proper function to adapt the representation consumer C to some aspect A0 of the environment, in particular by leading S to behave appropriately with respect to A0.
- (R3) There are (in addition to R0) transformations of R0, R1...Rn, that have as their function to adapt the representation consumer C to corresponding transformations of A0, A1...An.

According to the teleological definition, what some aspect of a system represents, indeed that it is a representation at all, *has nothing to do with what it is or is not coupled with*. Instead, an aspect of a system represents whatever state of affairs (or process) that the representation adapts the representation consuming part of the system to; that is, rather than representing what the world actually is (or isn't) like when the representation is tokened, the representation represents what the world would have to be like for the representation consumer to perform its function. Though, there is not time to defend it fully here (I have done that in Chemero 1998, Chapter 1), I will briefly point out a few of its features. First, as mentioned above, since it requires a representation to have functions, it is *teleological* (R2). Second, it requires that the representation serve as a representation in the context of producing and consuming devices (R1). Third, it has an explanation of (at least some kinds of) misrepresentation built in to it. Since the content of a representation is determined by its function, along with those of the representation producer and consumer, its content will remain constant even in cases in which one or

¹¹ The notion of ‘genic representation’ in Wheeler and Clark (1997) is essentially the same as the teleological notion. So Clark, it might seem, has in the space of just over a year gone from rejecting “decouplability” to thinking that it is a crucial part of a definition of representation, and back again. He has not though: in a footnote in Wheeler and Clark (1997), the distinction between weak (genic) and strong (decoupled) representation is re-stated.

¹² See also Millikan 1984, 1993; Bechtel 1998. Note that Millikan, at least, intends this definition to cover both mental and linguistic representations. I intend it to cover only the former.

more of the producer, consumer and representation itself fails to work properly. That is, just as the function of a sperm is to fertilize an egg, despite the fact that the number that do so is vanishingly small, so the function of an icon that represents “chicken-here-now” does not change, even in cases in which the icon is produced or used improperly. Fourth, it requires that a representation be part of a system of representations (R3). Fifth, it requires that we follow Millikan (1984) in focusing on the representation consumer in determining the content of a representation--the content is the way the world would need to be for the behavior caused by the representation consumer to be adaptive (R2).

Most important for our purposes, though, is that defining representation this way, rather than in terms of (de)coupling, avoids the ambiguity that attends the phrase ‘reliable presence’. Furthermore, the teleological definition is *not revisionist*: like the tradition, it treats representation as a generic term, allowing for both coupled and non-coupled species. It is a moderately liberal definition of representation--more restrictive than Wheeler’s which requires only the possibility of misrepresentation, but less so than Grush’s emulation theory. This moderate liberalism is especially important for the “New Computationalism” that we are here to formulate. If this “New Computationalism” is to be embodied, concrete, embedded, active, situated, and so on, it must embrace the extant cognitive science that is all of these things. Yet if ‘representation’ is defined in terms of “reliable presence,” and so decoupling, even most connectionist networks will not be representational, which means that they will also be non-computational. On the other hand, defining ‘representation’ teleologically, as I have suggested, makes it possible that those who have defined themselves in opposition to the older version of computationalism might view proponents of the “New Computationalism” as natural allies. This, I take it, is a good thing.

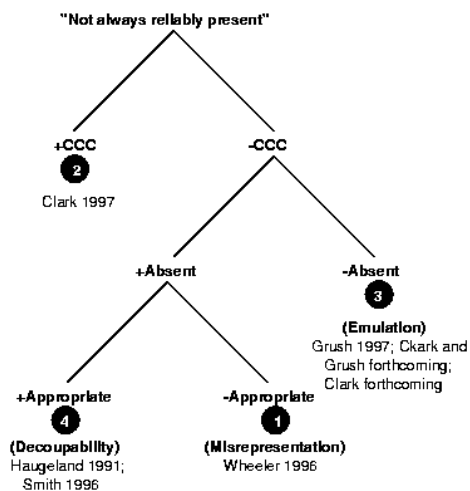


Figure 1

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